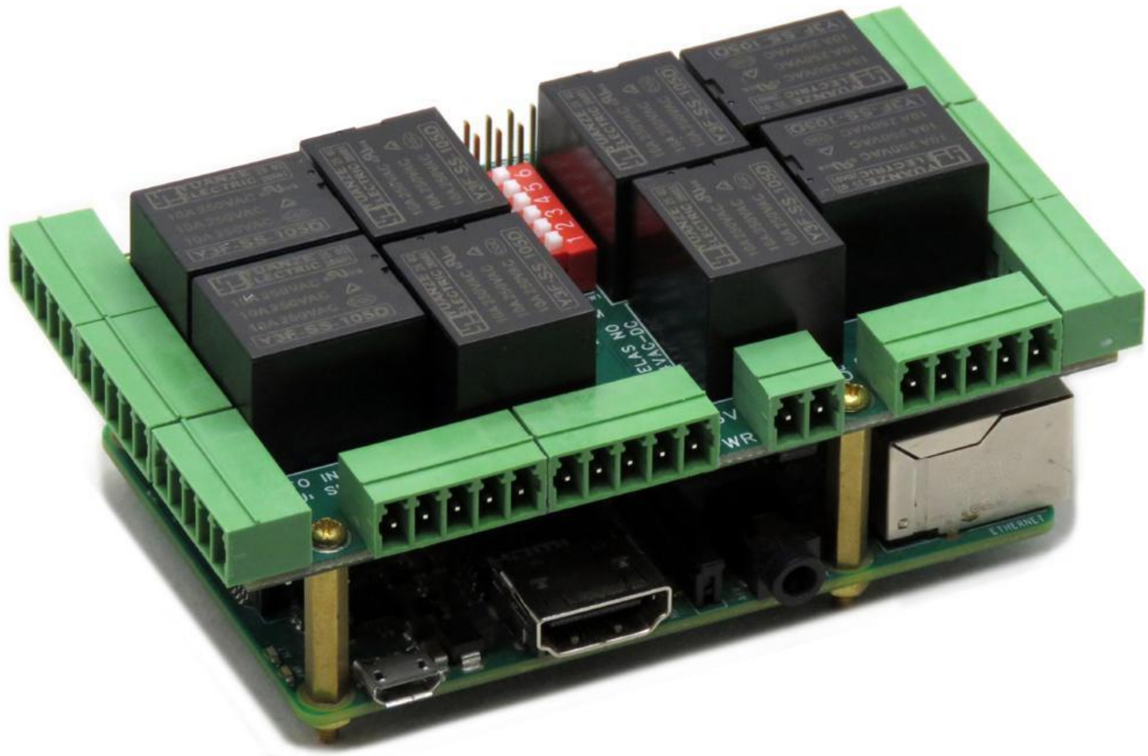
HOME AUTOMATION Card for Raspberry Pi

USER'S GUIDE VERSION 5

SequentMicrosystems.com

GENERAL DESCRIPTION



Compatible with all Raspberry Pi models from the Raspberry Pi Zero to Raspberry Pi 5, the Home Automation HAT provides a comprehensive and cost-effective platform for residential and light commercial automation projects. By combining analog inputs, relay outputs, digital inputs, PWM outputs, and 0–10 V analog outputs on a single card, it eliminates the need for multiple expansion boards and simplifies system integration.

The card can monitor temperatures in up to eight zones using its analog inputs, making it ideal for heating, ventilation, air conditioning (HVAC), and environmental monitoring applications. Eight onboard relays provide direct control of heating and cooling equipment, pumps, valves, fans, and other electrical loads. The system can be expanded further by stacking additional Sequent Microsystems cards, such as the Eight Relays Card for sprinkler systems or the Four Relays Card for controlling higher-voltage appliances and equipment.

Eight optically isolated digital inputs allow the card to interface safely with security sensors, door contacts, motion detectors, alarm systems, and other field devices. Optical isolation protects both the Raspberry Pi and connected equipment from electrical noise and voltage transients commonly encountered in building automation environments.

For lighting control applications, the card includes four PWM open-drain outputs capable of controlling external loads powered from supplies up to 24 VDC. These outputs can be used for LED lighting, low-voltage lighting systems, and other devices requiring pulse-width modulation control. In addition, four 0–10 V analog outputs provide compatibility with industry-standard lighting dimmers, variable-speed drives, dampers, valves, and other building automation equipment requiring analog control signals.

An integrated RS485/MODBUS RTU interface provides virtually unlimited expansion capabilities. The interface can operate in slave mode under Raspberry Pi control or allow the card to function as a standalone MODBUS RTU device connected directly to a PLC or other industrial controller, eliminating the need for a Raspberry Pi.

For increased system reliability, the card includes an onboard hardware watchdog that continuously monitors Raspberry Pi operation and can automatically power cycle the computer in the event of a software lockup or system failure. This feature is particularly valuable for unattended automation systems where maximum uptime is required.

Mechanically, the Home Automation HAT conforms to the Sequent Microsystems Modular Industrial format and is compatible with the free 3D-printable stackable enclosure available from our website. All cards in this family share the same mechanical dimensions, mounting provisions, connector locations, LED arrangement, and pushbutton position, enabling seamless integration and interchangeability.

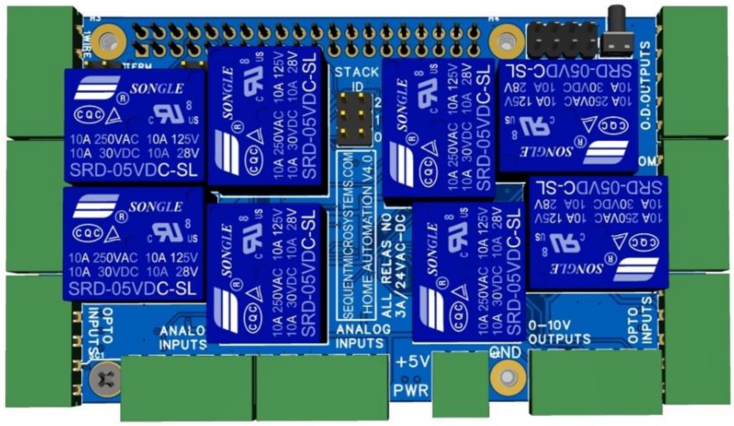
FEATURES

- Eight relays with status LEDs and Normally Open (N.O.) contacts
- Stack up to eight cards on a single Raspberry Pi
- Eight 12-bit analog inputs with 25 Hz sampling rate
- Four 13-bit analog outputs (0–10 V dimmer control)
- Four PWM open-drain outputs, up to 24 VDC and 4 A
- Eight optically isolated digital inputs
- Event and contact-closure counters up to 100 Hz
- Four quadrature encoder inputs
- 26 Raspberry Pi GPIOs remain available for user applications
- 1-Wire and RS485 communication ports
- Pluggable field-wiring connectors for 26–16 AWG wires
- Onboard hardware watchdog
- Onboard resettable fuse
- Reverse-polarity power supply protection
- 32-bit processor running at 64 MHz
- Hardware self-test using optional loopback cable
- Open-source hardware with published schematics
- All mounting hardware included: brass stand-offs, screws, and nuts

- Command line
- Python Library
- Node-Red Nodes
- Domoticz Plugin
- OpenPLC example integration
- Firmware Update

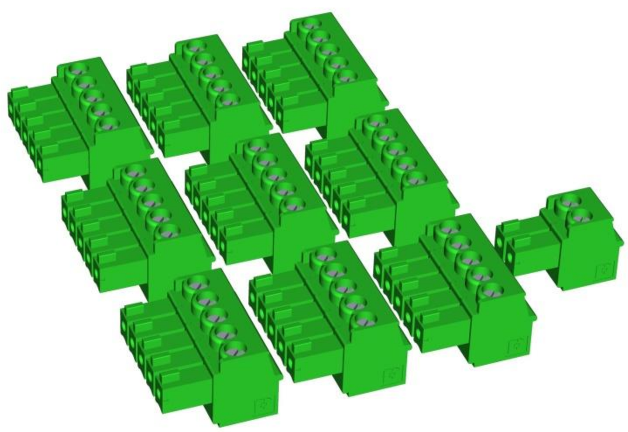
WHAT IS IN YOUR KIT

1. Home Automation Card for Raspberry Pi



2. Brass standoffs, screws and nuts

3. Connector plugs

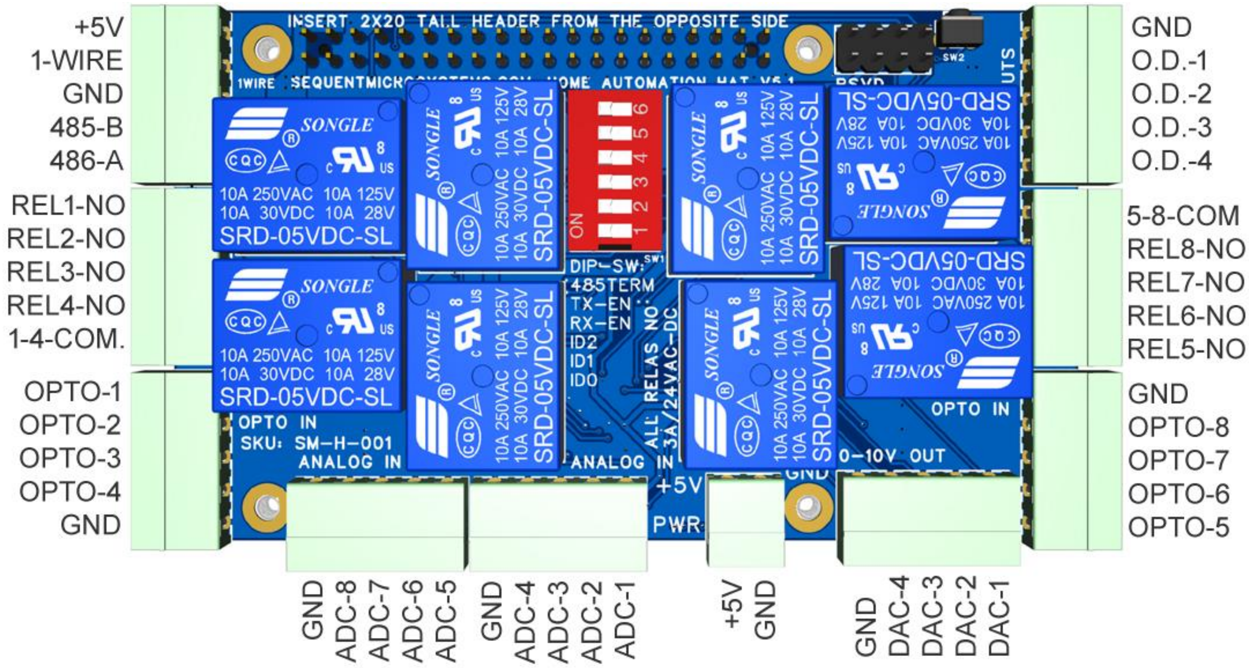


QUICK START-UP GUIDE

1. Plug your Home Automation Card on top of your Raspberry Pi and power up the system.
2. Enable I2C communication on Raspberry Pi using raspi-config.
3. Install the software from github.com:
    a. ~$ git clone https://github.com/SequentMicrosystems/ioplus-rpi.git
    b. ~$ cd /home/pi/ioplus-rpi
    c. ~/ioplus-rpi$ sudo make install
4. ~/ioplus-rpi$ ioplus

   The program will respond with a list of available commands.

BOARD LAYOUT



Your Home Automation Card is supplied with all required mounting hardware, including standoffs, screws, and nuts. Up to eight 8-RELAYS Cards can be stacked on a single Raspberry Pi, providing a total of 64 relay outputs.

Eight status LEDs indicate the state of the corresponding relays. An LED illuminates when its relay is energized. A separate power LED indicates that the board is powered.

The **General-Purpose Pushbutton (GPB)** is connected to Raspberry Pi GPIO26 (physical pin 37 on the 40-pin GPIO header). The pushbutton can be used to trigger user-defined actions such as safe shutdown, reboot, or application-specific functions. To use the pushbutton, a software script or application must be configured on the Raspberry Pi.

All relay contacts are brought out to pluggable field-wiring connectors. Each relay provides only **Normally Open (NO)** connections. Groups of four relays share the common pin.

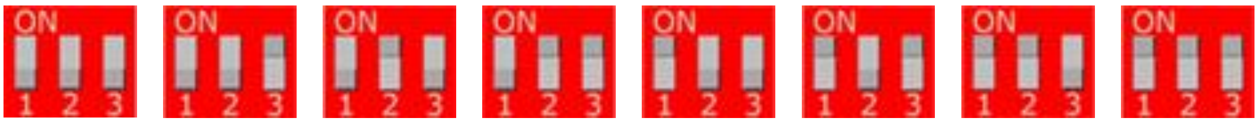
The configuration DIP-Switch SW1 is used to configure the card:

- The three leftmost switches select the card's stack level (see the next section).
- The rightmost switch enables the RS485 termination resistor. Set it ON when the card is the last device on the RS485 network.

The next two switches, 485-TX and 485-RX, connect the RS485 interface to the Raspberry Pi serial port. Set the switches OFF to enable communication through the onboard RS485 port. Set the switches ON if the Raspberry Pi serial port is being used to drive the serial port.

STACK LEVEL SETTINGS

Up to eight Home Automation Cards can be installed on top of one Raspberry Pi. Cards can be installed on Raspberry Pi in any order. The 3 leftmost positions of the DIP switch select the stack level of the card, as follows:



0	1	2	3	4	5	6	7

RS-485 COMMUNICATION

The card includes a standard RS485 transceiver that can be accessed either by the onboard processor or directly by the Raspberry Pi. The desired operating mode is selected using two configuration switches on the onboard DIP switch.

When the corresponding DIP switches are **ON**, the Raspberry Pi has direct access to the RS485 interface and can communicate with any RS485-compatible device. In this mode, the card functions as a passive hardware bridge, providing only the electrical interface required by the RS485 standard. To enable this configuration, the onboard processor must be instructed to release control of the RS485 bus.

When the DIP switches are **OFF**, the onboard processor takes control of the RS485 interface and the card operates as a **MODBUS RTU slave**. In this mode, any MODBUS master can read all input values and control all outputs using standard MODBUS RTU commands. A complete description of the implemented MODBUS registers and commands is available on GitHub:

https://github.com/SequentMicrosystems/smtc-rpi/blob/main/MODBUS.md

In both operating modes, the onboard processor must be configured to either release control of the RS485 interface (DIP switches ON) or manage the RS485 communications (DIP switches OFF). Refer to the command-line online help for detailed configuration instructions.

POWER REQUIREMENTS

The 8-RELAYS card requires +5V power, supplied either from the Raspberry Pi expansion bus, or from its own pluggable connector. The on-board relays are powered at +5V (See Schematic).

8-RELAYS Card current consumption:	10 mA @ +5V (all relays OFF)
	700 mA @ +5V (all relays ON)

The power connector which powers the 8-RELAYS card can supply up to 8A and is protected by a 7A resettable fuse. We recommend using a 5V regulated power supply rated at 5A or higher. The 8-RELAYS card can be stacked up to eight levels. A multi-stack configuration can be powered from any of the cards.

HARDWARE WATCHDOG

The Home Automation Card includes an onboard hardware watchdog designed to ensure reliable operation in unattended and mission-critical applications.

The watchdog is disabled at power-up and becomes active after receiving the first reset command from the Raspberry Pi. The default watchdog timeout is 120 seconds.

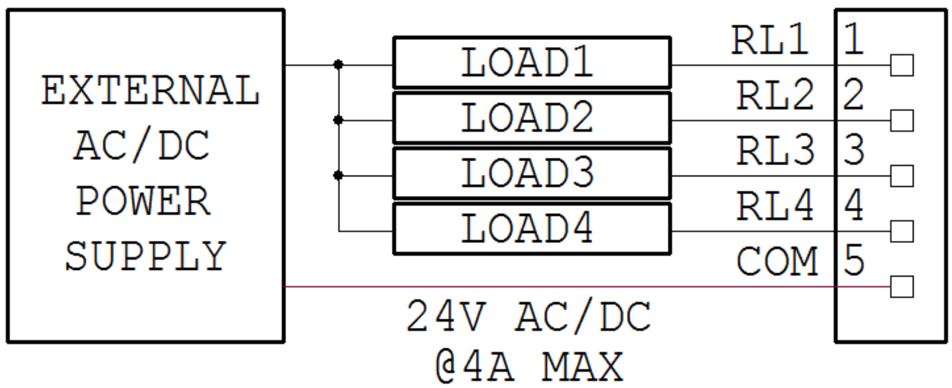
Once activated, the Raspberry Pi must periodically issue a reset command over the I²C interface before the watchdog timer expires. If no reset command is received within the configured timeout period, the watchdog removes power from the Raspberry Pi and automatically restores it after 10 seconds, forcing a complete system restart.

Both the startup timeout and the active watchdog timeout can be configured from the command line. The watchdog also maintains a non-volatile reset counter stored in onboard flash memory. The number of watchdog-triggered resets can be viewed or cleared using command-line commands.

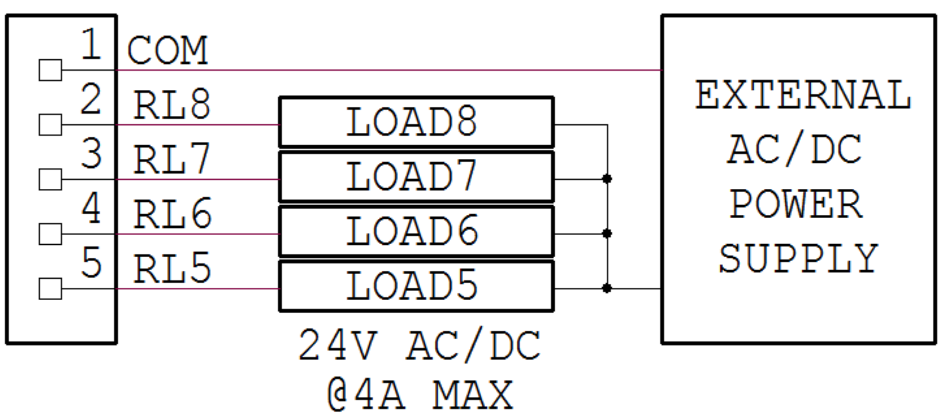
A complete description of all watchdog commands, status information, and configuration options is available through the built-in online help system.

NORMAL-OPEN RELAYS

Eight onboard relays are arranged in two groups of four and connected to two field-wiring connectors. Each connector provides one common terminal and four Normally Open (N.O.) relay contacts. Due to PCB trace width and spacing limitations, the relay contacts are rated for a maximum of 24 VAC/DC and 4 A per connector. The 4 A limit applies to the combined current of all loads connected to a given 4-relay connector. For example, each connector can switch one 4 A load, two 2 A loads, or four 1 A loads, provided that the total current does not exceed 4 A.



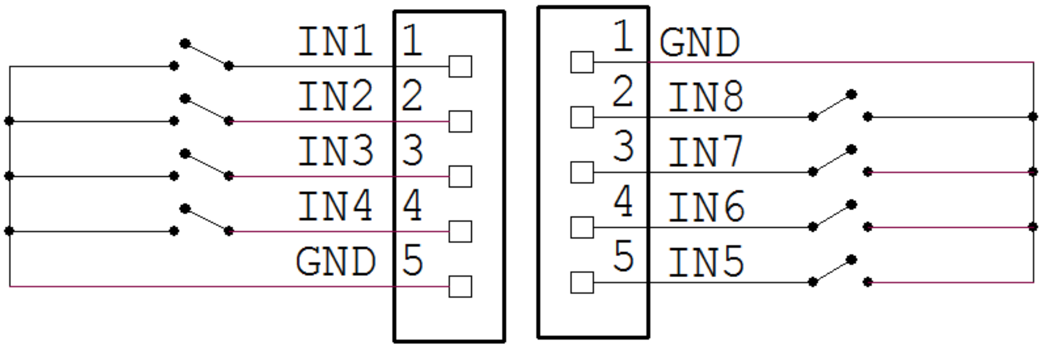
RELAYS 1-4 CONFIGURATION



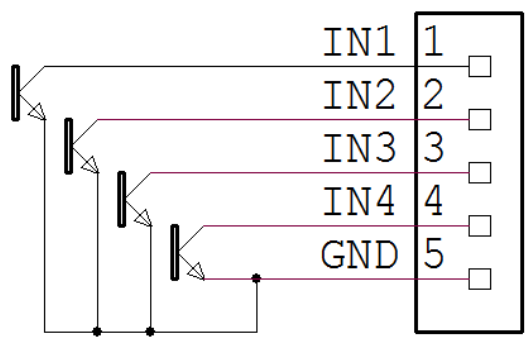
RELAYS 5-8 CONFIGURATION

OPTICALLY ISOLATED INPUTS

Each optically isolated input includes a 1 kΩ pull-up resistor connected to the 5 V supply. The inputs can be used to detect contact closures, interface with open-collector or open-drain outputs, or connect directly to quadrature encoders. This flexibility makes the inputs suitable for a wide range of sensing and monitoring applications, including switches, proximity sensors, pulse generators, flow meters, and rotary encoders.

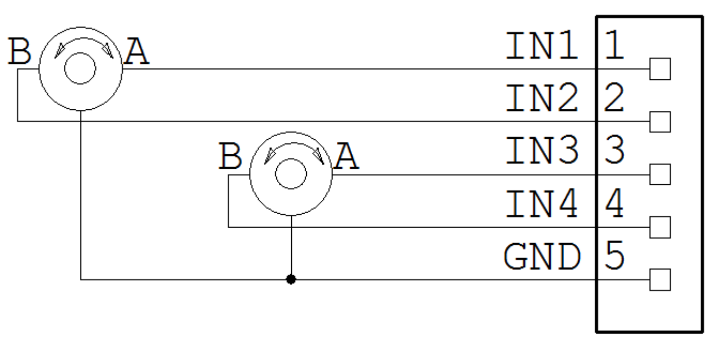


CONTACT CLOSURE CONFIGURATION



OPEN-DRAIN/OPEN-COLLECTOR CONFIGURATION

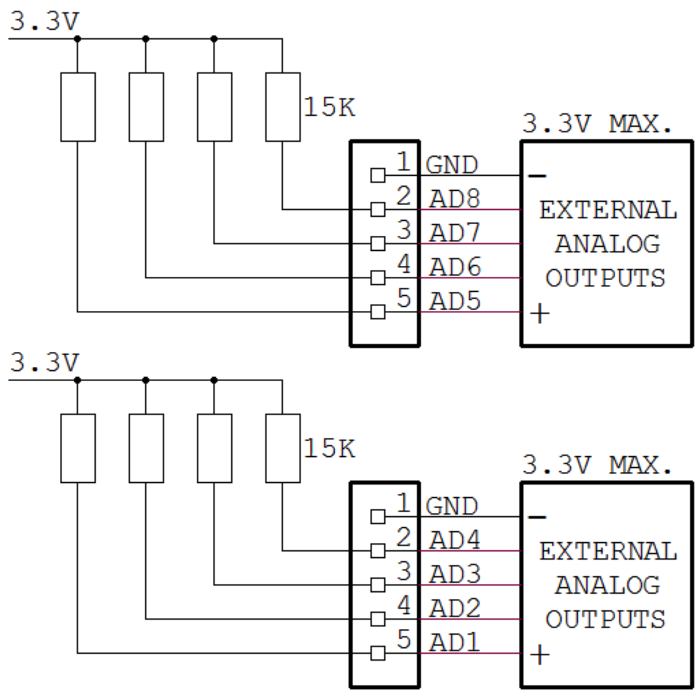
The Home Automation Card can count contact closure signals on the rising or falling edge of the signal, up to 100 Hz.



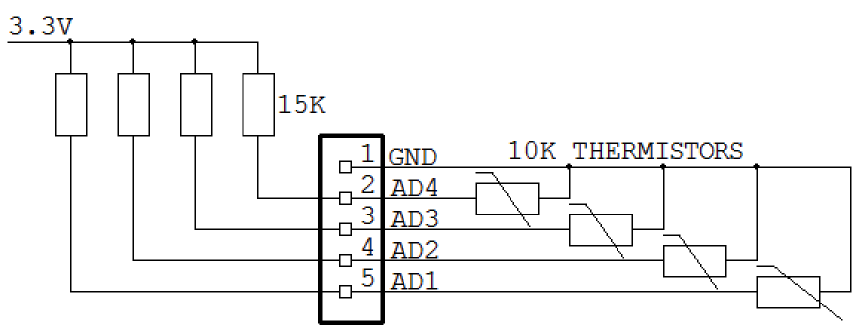
QUADRATURE ENCODER CONFIGURATION

0-3.3V ANALOG INPUTS

Eight analog inputs can measure signals in the 0-3.3V range. Each input has a 15Kohm resistor pullup. Using a 10K thermistor the analog inputs can be used to measure temperatures.



ANALOG INPUTS CONFIGURATION

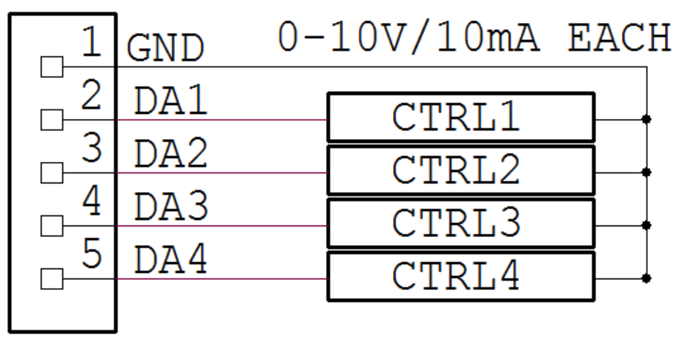


TEMPERATURE MEASUREMENT CONFIGURATION (4 OF 8 CHANNELS)

0-10V OUTPUTS/LIGHT DIMMERS

There are two widely used standards for 0–10 V dimming control: the IEC 60929 Annex E standard, which uses current-sinking control, and the ESTA E1.3 standard, which uses current-sourcing control. Both standards specify a maximum control current of 2 mA per channel. The Home Automation Card includes an onboard 12 V power supply and is fully compatible with both dimming methods.

Each of the four analog outputs can source up to 10 mA, allowing a single channel to control up to five standard 0–10 V dimmer interfaces. As a result, the card can control up to 20 dimmable lighting circuits when multiple dimmer controllers are connected. In addition to lighting applications, the 0–10 V outputs can be used to control industrial devices such as variable-speed drives, dampers, valves, and other equipment requiring a 0–10 V control signal with an input current of less than 10 mA per channel.



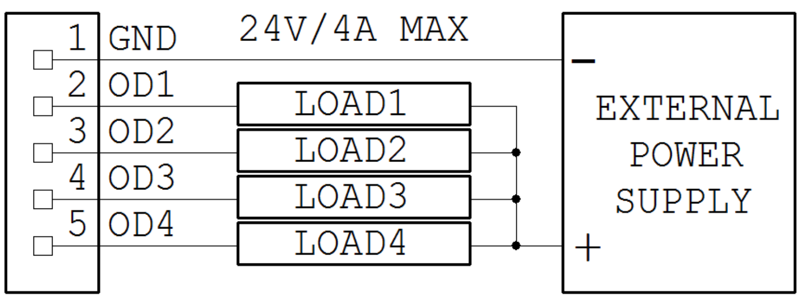
LIGHT DIMMERS CONFIGURATION

OPEN DRAIN OUTPUTS WITH PWM

The Home Automation Card provides four open-drain outputs, each capable of switching loads up to 4 A. An external power supply of up to 24 VDC is required to power the connected loads. The outputs can be controlled as simple on/off switches using the command interface or operated in proportional mode using Pulse Width Modulation (PWM).

The PWM frequency is fixed at 48 kHz, allowing smooth control of compatible loads such as LED lighting, DC motors, heaters, and other power devices. The duty cycle can be adjusted from 0% to 100%, providing precise output power control across the entire operating range.

The external power supply must be sized to provide the total current required by all connected loads. For example, if multiple outputs are used simultaneously, the power supply must be capable of delivering the sum of the currents drawn by all active loads.



OPEN DRAIN OUTPUTS CONFIGURATION

SPECIFICATIONS

ON BOARD RESETTABLE FUSE: 7A

OPEN DRAIN OUTPUTS:

- Maximum output current: 4A
- Maximum output voltage: 24V
- PWM frequency: 48KHz

ANALOG INPUTS:

- Maximum input voltage: 3.3V
- Input Impedance: 50 KΩ
- Resolution: 12 bits
- Sample rate: 250 samples/sec.

GPIO LINES:

- Directly from the on-board STM32F030 microprocessor under control of the software

DAC OUTPUTS:

- Resistive load: Minimum 1 KΩ
- Accuracy: ±1%

OPTO-ISOLATED INPUTS:

- Pull-up Resistor: 1K @ 5V
- Isolation Resistance: Minimum 10^{12} Ω

RELAY OUTPUTS

- Maximum current/voltage: 5A/48V

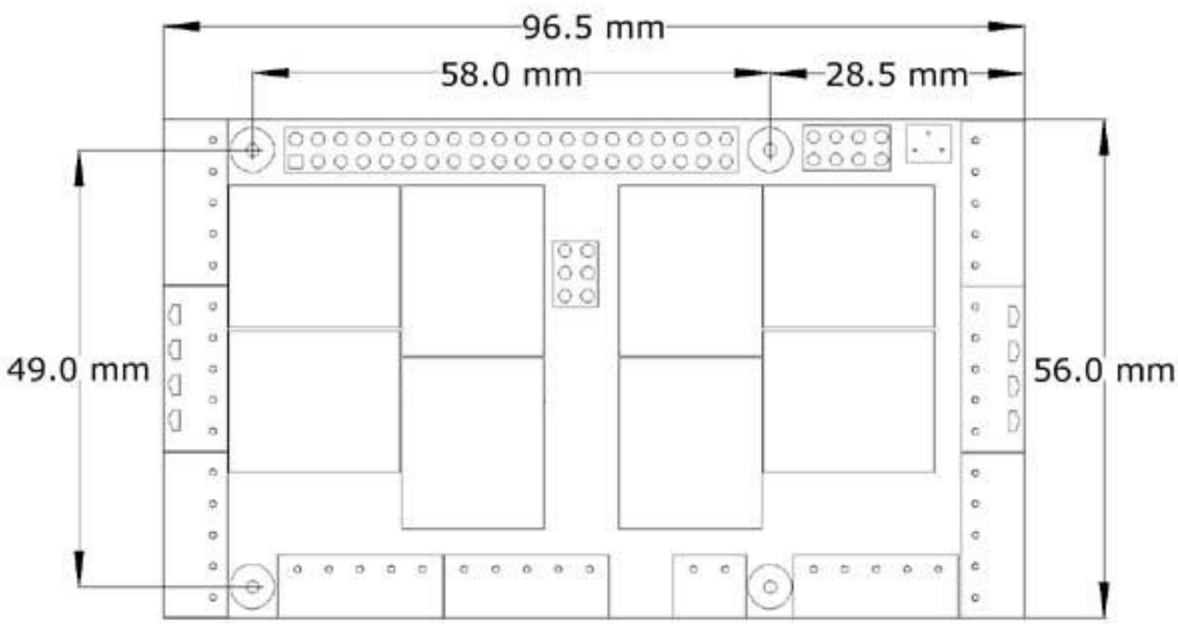
POWER CONSUMPTION:

- 50 mA @ +5V (all relays OFF)
- 750 mA @ +5V (all relays ON)

For detailed information on connecting devices to the various inputs and outputs of the Home Automation Card, refer to the schematics available on our website as well as the datasheets for the components implementing each interface. For example, information regarding the optically isolated inputs can be found in the datasheet for the **TLP291-4** optocoupler used on the board.

It is the user's responsibility to ensure that all input and output voltages, currents, and operating conditions remain within the limits specified in the relevant component datasheets and in this User's Guide. Exceeding these limits may result in improper operation or permanent damage to the card and connected equipment.

MECHANICAL SPECIFICATIONS



The Home Automation Card and its compatible expansion cards may be stacked in any order. Cards from other manufacturers may also be used in the stack, provided that they do not share the same I²C address as any of the Sequent Microsystems cards (see the **Stack Level Jumpers** section for details).

For best reliability, it is recommended that both the Raspberry Pi and all stacked expansion cards be powered from the same power supply. When supplying power through a stack of cards, the powered Home Automation Card should be installed closest to the Raspberry Pi to minimize voltage drop and ensure proper power distribution throughout the system.

SOFTWARE SETUP

1. Have your Raspberry Pi ready with the latest OS.

2. Enable I2C communication:

~$ sudo raspi-config

1. Change User Password	Change password for default user
2. Network Options	Configure network settings
3. Boot Options	Configure options for start-up
4. Localisation Options	Set up language and regional settings to match..
5. Interfacing Options	Configure connections to peripherals
6. Overclock	Configure overclocking for your Pi
7. Advanced Options	Configure advanced settings
8. Update	Update this tool to the latest version
9. About raspi-config	Information about this configuration

P1	Camera	Enable/Disable connection to the Raspberry Pi Camera
P2	SSH	Enable/Disable remote command line access to your Pi
P3	VNC	Enable/Disable graphical remote access to your Pi using...
P4	SPI	Enable/Disable automatic loading of SPI kernel module
P5	I2C	Enable/Disable automatic loading of I2C kernel module
P6	Serial	Enable/Disable shell and kernel messages to the serial port
P7	1-Wire	Enable/Disable one-wire interface
P8	Remote GPIO	Enable/Disable remote access to GPIO pins

3. Install the Home Automation software from github.com:

~$ git clone https://github.com/SequentMicrosystems/ioplus-rpi.git

5. ~$ cd /home/pi/ioplus-rpi

6. ~/ioplus-rpi$ sudo make install

7. ~/ioplus-rpi$ ioplus

The program will respond with a list of available commands.

Type "ioplus -h" for online help.

After installing the software, you can update it to the latest version with the commands:

1. ~$ cd /home/pi/ioplus-rpi

2. ~/ioplus-rpi$ git pull

3. ~/ioplus-rpi$ sudo make install

ANALOG INPUTS/OUTPUTS CALIBRATION

All analog inputs and outputs are factory calibrated to an accuracy of ±1%. The firmware also provides commands that allow users to recalibrate the board or optimize the calibration for specific application requirements. Each channel is calibrated using a two-point method, and the firmware performs linear interpolation between the stored calibration points. Calibration coefficients are stored in non-volatile onboard Flash memory and are automatically applied to all subsequent measurements and output settings.

For best accuracy, select two calibration points as close as possible to the lower and upper limits of the intended operating range. To calibrate an input channel, a known and accurate DC voltage source must be connected to the input. For example, when calibrating a 0–3.3 V input, an adjustable precision power supply may be used to generate the required reference voltages. To calibrate an output channel, issue a command to set the output to a known value, measure the actual output voltage with a calibrated instrument, and then issue the appropriate calibration command to store the measured value.

Because the calibration curve is assumed to be linear, only two calibration points are required for each channel. If an incorrect value is entered during the calibration process, a RESET command can be used to restore all channels in the corresponding calibration group to their factory calibration values. Calibration may then be repeated as needed.

The board can also be calibrated without an external voltage reference by first calibrating the analog outputs and then connecting those calibrated outputs to the corresponding analog inputs. This method provides a convenient way to verify and calibrate the entire analog subsystem using only onboard resources.

The following commands are available for calibration:

Apply 0.1V to Analog Inputs

CALIBRATE ANALOG INPUTS TO LOW LIMIT:

ioplus <stack> cuin <channel> 0.1

Apply 3.2V to Analog Inputs

CALIBRATE ANALOG INPUTS TO HIGH LIMIT:

ioplus <stack> cuin <channel> 3.2

RESET CALIBRATION OF ANALOG INPUTS:

ioplus <stack> rcuin

SET 0-10V OUTPUTS TO LOW LIMIT:

ioplus <stack> uout <channel> 0.5

CALIBRATE 0-10V LOW LIMIT:

ioplus <stack> cuout <channel> <actual_value>

SET 0-10V OUTPUTS TO HIGH LIMIT:

ioplus <stack> uout <channel> 9.5

CALIBRATE 0-10V HIGH LIMIT:

ioplus <stack> cuout <channel> <actual_value>

RESET CALIBRATION OF 0-10V OUTPUTS:

ioplus <stack> rcuout

HOME AUTOMATION CARD SELF TEST

The firmware includes built-in self-test functions that allow users to verify proper operation of the Home Automation Card. These tests should be performed immediately after power-up and only with all field-wiring connectors disconnected from the board.

Relay Self-Test

To test the relay outputs, run the following command:

ioplus <stack> reltest

The card will sequentially energize each relay in numerical order at 150 ms intervals and then de-energize them in the same sequence and at the same rate. The test repeats continuously until interrupted from the keyboard.

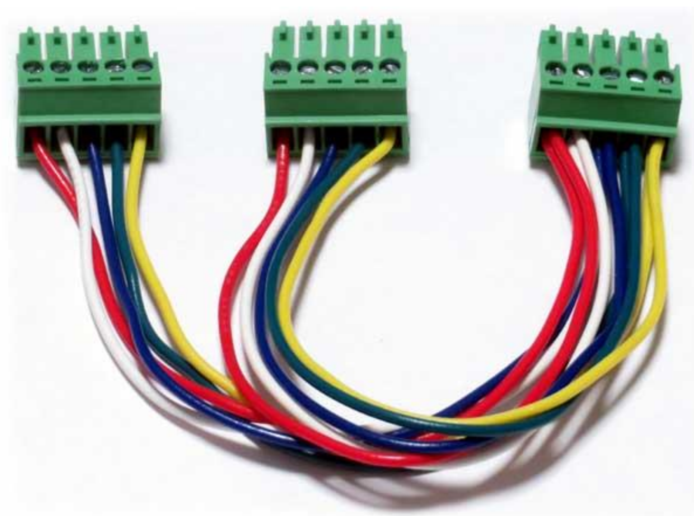
During the test, users can verify correct operation by listening for the relay clicks and observing the corresponding status LEDs as they turn on and off. Successful operation of all relays and LEDs indicates that the relay subsystem is functioning properly.

SELF TESTING USING THE LOOPBACK CABLE

All inputs, outputs, and relay contacts can be verified using a three-connector loopback cable, as shown in the figure below. The loopback cable connects the board's outputs to its corresponding inputs, allowing the firmware self-test routines to validate proper operation of the analog inputs, digital inputs, analog outputs, PWM outputs, and relay contacts.

A preassembled loopback cable is available from Sequent Microsystems. Alternatively, users can build their own cable using three of the pluggable connector plugs supplied with the card, provided that all connectors are not required for field wiring in the final installation.

Using the loopback cable together with the built-in self-test commands provides a quick and convenient method for verifying correct board operation after installation, maintenance, or troubleshooting.



Insert the loopback card into the IO Connector of the Home Automation Card and run the command:

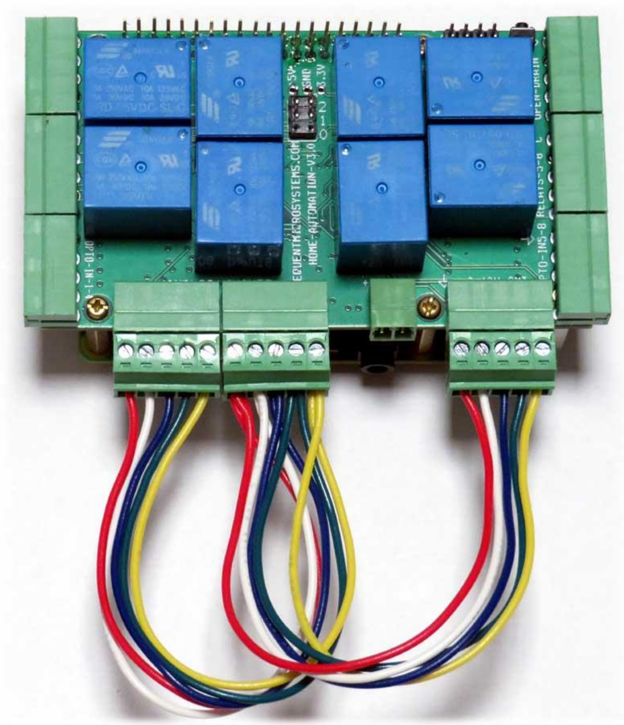
    ioplus <stack> iotest <test>

The optional <test> parameter indicates the number of the test to be run. If not specified, the software tries to find the cable and perform the required test. The parameter can have the values 1, 2 or 3.

SELF-TEST #1: In versions 4.0 and up, the GPIO pins on the upper left side connector have been replaced with an RS485 port and 1-Wire interface. Thus, the Self-Test #1 is no longer available.

SELF-TEST #2: Analog Inputs and Analog Outputs Verification.

Install the loopback cable as shown in the figure below. The connectors may be inserted in any order, as the test software automatically identifies the connected channels.



This test verifies the operation and calibration of the analog inputs and analog outputs by routing each analog output to its corresponding analog input through the loopback cable.

Run the following command:

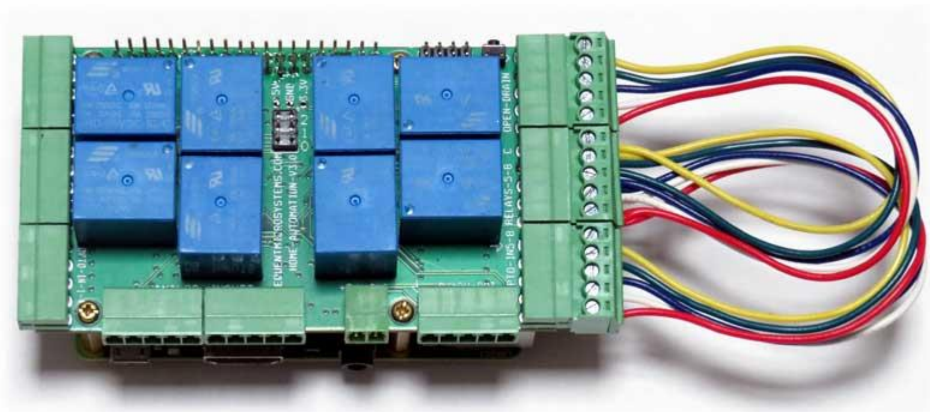
ioplus <stack> iotest 2

The firmware will generate a series of test voltages on the analog outputs and measure the corresponding values on the analog inputs. The measured results are compared against the expected values to verify proper operation of the analog subsystem. Any channels that fail the test will be reported on the console.

Successful completion of this test confirms correct operation of the analog outputs, analog inputs, associated signal paths, and calibration data stored in the board.

Self-Test #3: Open-Drain Outputs and Opto-Isolated Inputs Verification

Install the loopback cable as shown in the figure below. The connectors may be inserted in any order, as the test software automatically detects the connected channels.



This test verifies the operation of the four open-drain outputs and opto-isolated inputs 5 through 8 by using relays 5 through 8 to route the test signals. During the test, the firmware activates each open-drain output in sequence and monitors the corresponding opto-isolated input to confirm proper operation.

Run the following command:

ioplus <stack> iotest 3

The firmware will automatically cycle through all channels involved in the test and report the results on the console. Successful completion confirms proper operation of the open-drain outputs, relays 5–8, opto-isolated inputs 5–8, and the associated signal paths. Any detected failures will be identified and reported to assist with troubleshooting.